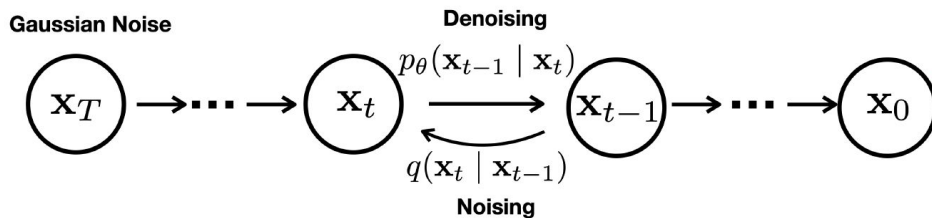


# Language Diffusion Models

# Background: Diffusion Model



## Forward process / Diffusion Process

$$q(\mathbf{x}_{1:T} | \mathbf{x}_0) := \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1}), \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

## Reverse Process

$$p_\theta(\mathbf{x}_{0:T}) := p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$$

## ELBO

$$\max_{\theta} \sum_i \ln p_\theta(x_i)$$

$$\int p_\theta(x|z)p(z)dz = \mathbb{E}_{z \sim q_\phi(\cdot|x)} \left[ \frac{p_\theta(x, z)}{q_\phi(z|x)} \right]$$

$$\ln p_\theta(x) = \ln \mathbb{E}_{z \sim q_\phi(\cdot|x)} \left[ \frac{p_\theta(x, z)}{q_\phi(z|x)} \right] \geq \mathbb{E}_{z \sim q_\phi(\cdot|x)} \left[ \ln \frac{p_\theta(x, z)}{q_\phi(z|x)} \right]$$

# Two Paths towards LDM

## Continuous-latent-space Diffusion/Reverse Process

### Diffusion-LM Improves Controllable Text Generation

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## Discrete-latent-space Diffusion/Reverse Process

ByteDance | Seed

AIR 清华大学 人工智能研究院  
GenSi

### Seed Diffusion: A Large-Scale Diffusion Language Model with High-Speed Inference

<sup>1</sup>ByteDance Seed <sup>2</sup>Institute for AI Industry Research (AIR), Tsinghua University  
<sup>3</sup>SIA-Lab of Tsinghua AIR and ByteDance Seed

#### Abstract

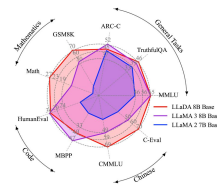
We present **Seed Diffusion Preview**, a large-scale language model based on discrete-state diffusion, offering remarkably fast inference speed. Thanks to non-sequential, parallel generation, discrete diffusion models provide a notable speedup to mitigate the inherent latency of token-by-token decoding, as demonstrated recently (e.g., Mercury Coder [1], Gemini Diffusion [2]). Seed Diffusion Preview achieves an inference speed of **2.16k tokens/s** over H20 GPUs while maintaining competitive performance across a sweep of standard code evaluation benchmarks, significantly faster than contemporary Mercury and Gemini, establishing new state of the art on the speed-quality Pareto frontier for code models. Demo is available at [https://studio.seed.ai/exp/seed\\_diffusion/](https://studio.seed.ai/exp/seed_diffusion/).

#### Large Language Diffusion Models

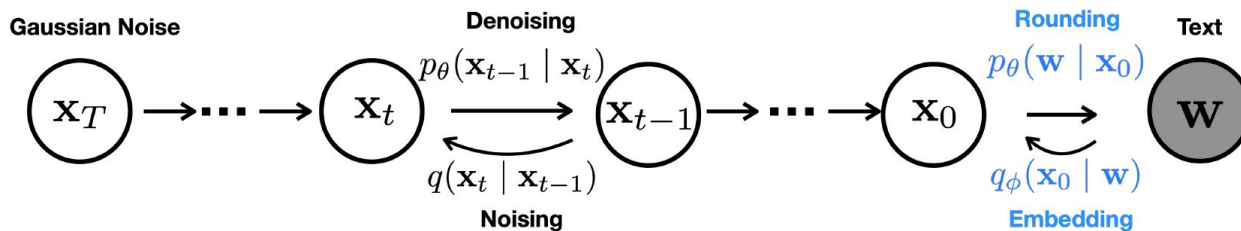
engqi Zhu<sup>1,†</sup> Zebin You<sup>1,†</sup> Xiaolu Zhang<sup>2,†</sup> Jingsyao Ou<sup>1</sup> Jun Hu<sup>2,†</sup> Jun Zhou<sup>2</sup>  
Yankai Lin<sup>1,†</sup> Ji-Rong Wen<sup>1</sup> Chongxuan Li<sup>1,†,§</sup>

#### Abstract

Autoregressive models (ARMs) are widely regarded as the cornerstone of large language models (LLMs). We challenge this notion by introducing **LLaDA**, a diffusion model trained from scratch under the pre-training and supervised fine-tuning (SFT) paradigm. LLaDA models distributions through a forward data masking process and a reverse process, parameterized by a vanilla Transformer to predict masked tokens. By optimizing a likelihood bound, it provides a principled generative approach for probabilistic inference. Across extensive benchmarks, LLaDA demonstrates strong *scalability*, outperforming our self-constructed ARM baselines. Remark-



# Continuous Latent Space Diffusion / Reverse Process

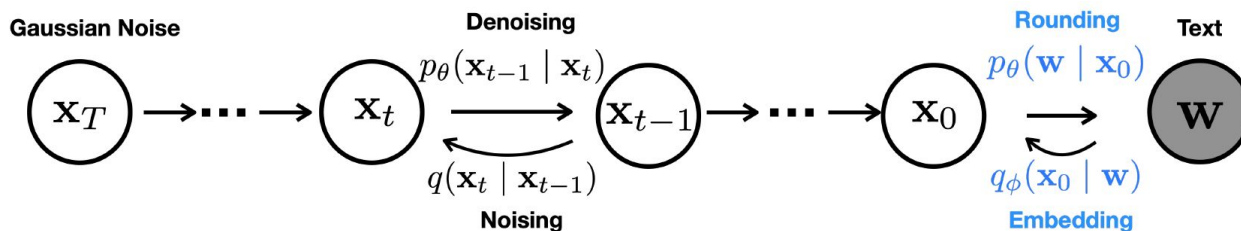


$$\mathcal{L}_{\text{vib}}(\mathbf{x}_0) = \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[ \log \frac{q(\mathbf{x}_T | \mathbf{x}_0)}{p_\theta(\mathbf{x}_T)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0, \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1) \right]$$

$$\mathcal{L}_{\text{vib}}^{\text{e2e}}(\mathbf{w}) = \mathbb{E}_{q_\phi(\mathbf{x}_0 | \mathbf{w})} [\mathcal{L}_{\text{vib}}(\mathbf{x}_0) + \log q_\phi(\mathbf{x}_0 | \mathbf{w}) - \log p_\theta(\mathbf{w} | \mathbf{x}_0)]$$

$$\mathcal{L}_{\text{simple}}(\mathbf{x}_0) = \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \|\mu_\theta(\mathbf{x}_t, t) - \hat{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2$$

# Continuous Latent Space Diffusion / Reverse Process



$$\mathcal{L}_{\text{vlb}}(\mathbf{x}_0) = \mathbb{E}_{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[ \log \frac{q(\mathbf{x}_T | \mathbf{x}_0)}{p_\theta(\mathbf{x}_T)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0, \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1) \right]$$

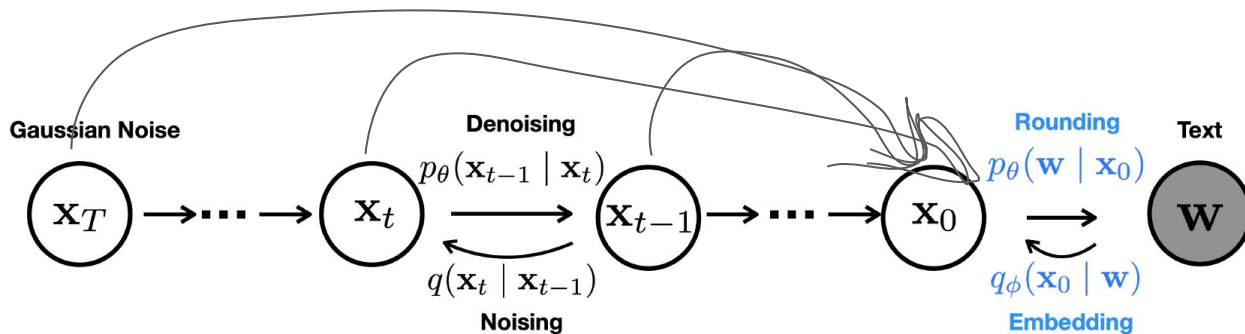
$$\mathcal{L}_{\text{vlb}}^{\text{e2e}}(\mathbf{w}) = \mathbb{E}_{q_\phi(\mathbf{x}_0 | \mathbf{w})} [\mathcal{L}_{\text{vlb}}(\mathbf{x}_0) + \log q_\phi(\mathbf{x}_0 | \mathbf{w}) - \log p_\theta(\mathbf{w} | \mathbf{x}_0)]$$

$$\mathcal{L}_{\text{simple}}(\mathbf{x}_0) = \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \|\mu_\theta(\mathbf{x}_t, t) - \hat{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2$$

**Challenge:** Empirically, the model fails to estimate  $\mathbf{x}_0$  that commits to words

**Their Interpretation:** Insufficient Emphasis in the objective function

# Continuous Latent Space Diffusion / Reverse Process



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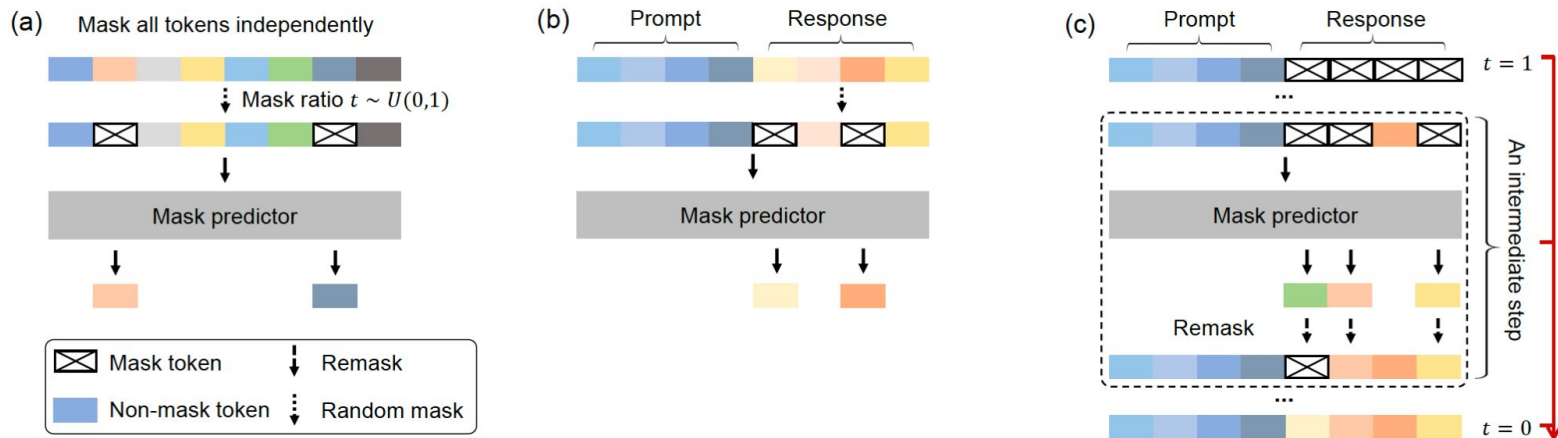
**Their solution:**

$$\mathcal{L}_{\mathbf{x}_0\text{-simple}}^{\text{e2e}}(\mathbf{x}_0) = \sum_{t=1}^T \mathbb{E}_{\mathbf{x}_t} \|\mathbf{f}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|^2 \quad \mathbf{x}_0 \text{ parametrization}$$

+

*clamping trick*

# Discrete Latent Space Diffusion / Reverse Process



$$-\mathbb{E}_{l, r_0, r_l} \left[ \frac{L}{l} \sum_{i=1}^L \mathbf{1}[r_l^i = \mathbf{M}] \log p_{\theta}(r_0^i | p_0, r_l) \right]$$

# Design Choices: On-Policy RL for Efficient Decoding

**Challenge:** efficient decoding?

**Their solution:**

$$\mathbb{E}_{\substack{\text{prompt} \sim p_{\text{data}} \\ \tau \sim p_{\theta}(\cdot | \text{prompt})}} [|\tau| - V(\tau[0])]$$

$$|\tau| \propto \mathbb{E}_{i,j \in \{0, \dots, K\}} \frac{1}{d_{\text{Lev}}(\tau[i], \tau[j])}$$

**Levenshtein distance,**  $d_{\text{Lev}}(\mathbf{x}_a, \mathbf{x}_b)$

# Limitations: Equivalence between AutoRegressive Model and Diffusion Model

An important perspective to understand the essentials of mask-based diffusion models is its equivalence to any-order autoregressive modeling, which has been revealed and illustrated by [22]. This perspective builds the correlation between ELBO and an expected log-likelihood of any-order autoregressive models as:

$$\mathcal{J}(\theta) = -\mathbb{E}_{\pi \sim U(S_d)} \left[ \sum_{r=1}^{d=|\mathbf{x}|} \log p_{\theta}(\mathbf{x}_{\pi(r)} \mid \mathbf{x}_{\pi(<r)}) \right] \quad (7)$$

Here  $S_d$  stands for the symmetric group of all possible permutations  $\pi$  over  $\{0, 1, \dots, d-1\}$ .  $p_{\theta}(\mathbf{x}_{\pi(r)} \mid \mathbf{x}_{\pi(<r)})$  models the conditional probability of  $\mathbf{x}_{\pi(r)}$  given all preceding tokens in the permutations  $\pi$  denoted as  $\mathbf{x}_{\pi(<r)}$ .

**Challenge:** Redundant, detrimental, or misaligned with the natural of language

**Their solution:** Distillation?

# Connections with LLMs

1. BERT is a one-step diffusion model
2. ENTP is diffusion model with deterministic masking design